



Day 5 Task - Disk, Partition & File System Examination



Trios cyber • 7 Sept (Edited 9 Sept)

100/100 | Due 8 Sept

Practical Assignment: Disk, Partition & File System Examination

Dear Students,

For this practical assignment, you are required to examine the storage structure of a digital device or forensic image and identify how data is organized within its partitions and file system.

The objective of this exercise is to understand disks, partitions, allocated and unallocated space, slack space, and common file systems used in digital forensics.

Part 1: Examine the Storage Structure

Using the forensic image created in Task 1 or a new disk image, examine the available storage information.

Where available, identify:

- Disk/storage source
- Disk size
- Partitions
- Partition size
- Partition type
- File-system type

Take appropriate screenshots of the storage structure.

Part 2: Examine the File System

Identify the file system present in the image, where available.

Research and explain the basic structure of the identified file system.

In your report, explain the relevant characteristics of:

- NTFS
- FAT
- EXT3/EXT4



Include the following:

- How files/directories are organized
- Important file-system structures
- How metadata is associated with files
- How the file system records the location of data

Part 3: Allocated, Unallocated & Slack Space

Explain and identify, where applicable:

- Allocated space
- Unallocated space
- Slack space

For each, explain:

- What it means
- Where it occurs
- What type of information it may contain
- Why it is relevant in digital forensics

Part 4: Directory & File Structure

Navigate through the available directories and files.

Select at least 5 files and document:

- Filename
- Full path
- File size
- File-system location/information, where available
- Relevant timestamps
- File attributes, where available

Take screenshots showing the directory and file structure.

Part 5: Hidden or Unexpected Storage

Examine the available storage structure for any unexpected partitions, directories or storage areas.

If none are present or accessible, state:

“Not Available/Not Observed.”

Do not attempt to modify the image or create hidden data.

Part 6: Document Your Work

Prepare a PDF or DOCX report containing:

1. Student Information



2. Objective
3. Disk/storage information
4. Partition information, where available
5. File-system identification
6. NTFS, FAT and EXT comparison
7. Allocated, unallocated and slack-space explanation
8. Directory/file examination
9. Screenshots
10. Observations
11. Conclusion

Submission Requirements

Submit ONE file only:

- PDF (.pdf) OR
- Microsoft Word document (.docx)

Important Notes

Use only the forensic image or evidence source provided/authorized for the exercise.

Do not modify the forensic image.

If a particular storage structure is not available in your image, clearly mention “Not Available.”

Expected Learning Outcome

After completing this practical, you should be able to:

1. Identify basic disk and partition structures.
2. Recognize common file systems.
3. Explain allocated, unallocated and slack space.
4. Examine directory and file structures.
5. Understand why file-system structures are important in forensic examination.
6. Identify potentially relevant storage areas for further examination.

Submission: Upload your completed PDF/DOCX report to this Google Classroom assignment before the specified deadline. Late submissions are allowed only till after 48 hours from the time the task was posted but it may result in deduction of marks.

Please read the instructions carefully before beginning the assignment.

Your work

Marked



Day-5.pdf
PDF



+ Add or create

Resubmit

Your teacher is not accepting work at this time



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